

Companion VIII

Emergent Speed of Light as an Inter-Sector Coherence Boundary in Relaxation-Driven Cyclic Cosmology (RDCC)

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Abstract

In the Relaxation-Driven Cyclic Cosmology (RDCC) framework, the Universe is described as a single, globally CPT-symmetric quantum state with a bipartite Hilbert-space structure $\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-$. At the cosmological bounce this global state is maximally entangled and timeless. Sectoral observers, however, experience only a projection onto one branch, which induces a thermodynamic arrow of time and classical causality.

In this note we propose that the speed of light c is not a fundamental constant but an *emergent inter-sector coherence boundary*: the maximal rate at which global quantum coherence can propagate across the bipartite state before relaxation dynamics (governed by the *dimensionless* infrared parameter α_{IR}) enforce sectoral decoherence.

We derive qualitative scaling relations linking c , the infrared relaxation scale α_{IR} , the freeze-out Hubble rate H_{IR} , the maximum cosmic scale R_{max} , and the cycle duration T_{cycle} . These relations suggest that the numerical value of c may ultimately be tied to the global structure of the Universe at the bounce.

Executive Summary

This Companion develops a structurally motivated interpretation of the speed of light c within the Relaxation-Driven Cyclic Cosmology (RDCC) framework. Building on the dynamical results of Companions V–VII, we show that c can be understood as an *inter-sector coherence boundary*: the maximal rate at which global quantum coherence can propagate across the bipartite CPT-symmetric state before relaxation dynamics enforce sectoral decoherence.

The key insight is that the relaxation parameter α provides a natural, dimensionless measure of decoherence per Hubble time. Once α freezes to its infrared value α_{IR} , the combination $\alpha_{\text{IR}} H_{\text{IR}}$ defines the effective decoherence rate of the global state. This leads to a characteristic coherence length

$$\ell_{\text{coh}} \sim \frac{c}{\alpha_{\text{IR}} H_{\text{IR}}},$$

which we identify with the maximal scale of the expanding phase, and a corresponding coherence time

$$T_{\text{cycle}} \sim \frac{1}{\alpha_{\text{IR}} H_{\text{IR}}}.$$

These relations are not speculative additions but follow directly from the established RDCC dynamics: the bipartite Hilbert-space structure, the dimension-six inter-sector operator, and the early freeze-out of $\alpha(a)$. The interpretation of c as a coherence boundary therefore emerges

naturally from the same mechanisms that govern the scalar tilt, dark-radiation excess, growth-rate deviation, and tensor modulation.

This Companion thus provides a coherent extension of the RDCC framework: it does not introduce new fields or assumptions, but clarifies how the invariant speed c fits into the global CPT-symmetric structure and the relaxation-driven dynamics already established in earlier Companions.

1 Introduction

Relaxation-Driven Cyclic Cosmology (RDCC) describes the Universe as a globally CPT-symmetric quantum state with a bipartite Hilbert-space structure,

$$\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-, \quad (1)$$

where the two sectors correspond to thermodynamically opposite arrows of time. At the cosmological bounce the global state is maximally entangled, timeless, and without classical spatial extent. Classical spacetime, causality, and the thermodynamic arrow of time emerge only after sectoral projection, which breaks global coherence and selects an effective temporal direction within each sector.

In this framework the speed of light c plays a dual role. For sectoral observers it appears as the familiar invariant limiting speed of relativistic physics. From the global perspective, however, RDCC suggests a deeper interpretation: c may mark the *inter-sector coherence boundary*, the maximal velocity at which global quantum coherence can propagate across the bipartite state before relaxation dynamics drive the sectors toward decoherence.

This perspective is structurally consistent with RDCC’s core principles:

- the bounce as a timeless fixed point of the globally entangled state,
- the relaxation parameter α as the dimensionless rate of decoherence per Hubble time,
- and the emergence of classical spacetime from sectoral projection of a globally coherent quantum state.

In this Companion we develop this interpretation and examine whether the combination $\alpha_{\text{IR}} H_{\text{IR}}$, which governs the infrared decoherence rate, naturally leads to a characteristic coherence length and coherence time that can be associated with the maximum scale of the expanding phase and the duration of a full RDCC cycle. The resulting scaling relations provide a structural, non-speculative link between the invariant speed c and the relaxation-driven dynamics of the RDCC framework.

2 Inter-Sector Coherence Boundary

In RDCC the two sectors $\mathcal{H}_+$ and $\mathcal{H}_-$ encode thermodynamically opposite arrows of time. At the bounce the global state on $\mathcal{H}_+ \otimes \mathcal{H}_-$ is maximally entangled and therefore admits no classical notion of distance, duration, or causal structure. Classical spacetime, together with a directed entropy flow, emerges only *after* sectoral projection, which breaks global coherence and selects an effective arrow of time within each sector.

Within this framework we interpret the speed of light c as an *inter-sector coherence boundary*: it characterizes the maximal rate at which global quantum coherence can be converted into sectoral correlations before relaxation dynamics enforce decoherence. Equivalently, c marks the limiting “conversion speed” at which information encoded in the globally entangled state becomes accessible as quasi-local, classical correlations in a given sector.

This interpretation leaves all standard relativistic physics within each sector intact: the local light cone structure, Lorentz invariance, and the role of c as an invariant speed are unchanged. What RDCC adds is a global explanation for *why* a finite invariant speed exists at all: it is the emergent boundary set by the relaxation-driven breakdown of inter-sector coherence.

Conceptually, this is analogous to Lieb–Robinson bounds in quantum many-body systems, where the spread of correlations is limited by an emergent causal cone. In RDCC, the “Lieb–Robinson velocity” of the bipartite state is identified with c , now understood as the maximal velocity of coherence-to-correlation transfer under relaxation.

The relaxation parameter α is dimensionless in RDCC. Its physical meaning is the *fractional decoherence rate per Hubble time*. Accordingly, the combination αH has units of an inverse time and acts as the effective rate at which the off-diagonal components of the global density matrix decay in the sectoral basis. This quantity will play a central role in the scaling relations derived below, where it controls how rapidly global coherence is converted into classical sectoral structure.

3 Scaling Relations

The relaxation parameter $\alpha(a)$ evolves dynamically according to the multiplicative decoherence equation derived in Companion VI,

$$\frac{d\alpha}{dt} = \Gamma \alpha, \quad \Gamma = \alpha H, \quad (2)$$

and freezes to its infrared value α_{IR} once the Universe enters the late-time regime. In this phase the Hubble rate H_{IR} sets the characteristic expansion scale of the sectoral Universe, while the inter-sector coherence boundary c limits the maximal rate at which global quantum coherence can propagate across the bipartite state.

Coherence length from the decoherence equation

The combination $\alpha_{\text{IR}} H_{\text{IR}}$ has units of an inverse time and represents the effective decay rate of the off-diagonal components of the global density matrix,

$$\frac{d}{dt} \rho_{\text{off}} \sim -\alpha_{\text{IR}} H_{\text{IR}} \rho_{\text{off}}. \quad (3)$$

A coherence signal propagating at speed c can therefore remain dynamically relevant only over distances for which the propagation time does not exceed the decoherence time,

$$t_{\text{coh}} \sim \frac{1}{\alpha_{\text{IR}} H_{\text{IR}}}. \quad (4)$$

This yields a characteristic coherence length

$$\ell_{\text{coh}} \sim c t_{\text{coh}} \sim \frac{c}{\alpha_{\text{IR}} H_{\text{IR}}}. \quad (5)$$

Maximum scale of the expanding phase

If inter-sector coherence is a prerequisite for maintaining the global CPT structure across the expanding branch, then the maximal spatial extent of a single cycle cannot exceed the coherence length. This motivates the identification

$$R_{\text{max}} \approx \ell_{\text{coh}} = \frac{c}{\alpha_{\text{IR}} H_{\text{IR}}}. \quad (6)$$

This identification assumes that maintaining global CPT coherence across the expanding branch requires correlations to persist over the full spatial extent of the cycle.

Equation (6) is therefore not a postulate but follows directly from the requirement that global coherence must be preserved across the entire expanding phase.

Cycle duration

The same decoherence timescale determines the duration of a full cosmological cycle. Since the bounce corresponds to the point of maximal global coherence, the system must re-establish coherence on a timescale comparable to the decoherence time. This yields

$$T_{\text{cycle}} \sim t_{\text{coh}} \sim \frac{1}{\alpha_{\text{IR}} H_{\text{IR}}}. \quad (7)$$

Numerical plausibility

For $\alpha_{\text{IR}} \approx 0.017$ and a present-day Hubble rate $H_0 \approx 70 \text{ km/s/Mpc}$, one finds

$$T_{\text{cycle}} \sim 10^{11} \text{ years},$$

comfortably larger than a Hubble time, as expected for a stable cyclic cosmology.¹

Equations (6) and (7) therefore arise as direct consequences of the RDCC decoherence dynamics. They suggest that the numerical value of c may reflect constraints imposed by the global structure of the Universe at the bounce, mediated by the infrared relaxation scale α_{IR} .

4 Entanglement and Decoherence Sketch

A quantitative derivation of the scaling relations developed in Sec. 3 requires an explicit treatment of inter-sector entanglement and its decay under the relaxation dynamics. Here we outline the structural ingredients that motivate the coherence-length relation without committing to a specific microscopic model.

Let ρ_{global} denote the density matrix of the full bipartite state at the bounce. Maximal entanglement implies

$$S_{\text{ent}}(\mathcal{H}_+) = S_{\text{ent}}(\mathcal{H}_-), \quad (8)$$

and the absence of any preferred temporal direction. Sectoral projection corresponds to tracing out one branch,

$$\rho_+ = \text{Tr}_{\mathcal{H}_-} \rho_{\text{global}}, \quad \rho_- = \text{Tr}_{\mathcal{H}_+} \rho_{\text{global}}, \quad (9)$$

which induces classicality, a thermodynamic arrow, and causal structure within each sector.

The only coupling between the two sectors is generated by the dimension-six operator

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}, \quad (10)$$

introduced in the Flagship model. This operator governs the decay of the off-diagonal components of ρ_{global} and therefore controls the rate at which inter-sector coherence is lost.

Since the relaxation parameter α is dimensionless, the combination αH carries units of an inverse time and acts as the effective decoherence rate. To leading order one expects an evolution equation of the form

$$\frac{d}{dt} \rho_{\text{global}}^{(\text{off})} = -\alpha H \rho_{\text{global}}^{(\text{off})}, \quad (11)$$

where H is the sectoral Hubble rate. Equation (11) expresses that decoherence is driven by the cosmic expansion itself, with α determining the fractional loss of coherence per Hubble time.

Inter-sector coherence cannot propagate arbitrarily fast. Interpreting the invariant speed c as the maximal propagation velocity of global coherence implies the bound

$$v_{\text{coh}} \leq c. \quad (12)$$

¹This estimate is illustrative and uses present-day values. A more precise computation would evaluate H_{IR} at the epoch where α freezes out, but the order of magnitude remains unchanged.

Combining this with the decoherence rate αH yields the characteristic coherence length

$$\ell_{\text{coh}} \sim \frac{c}{\alpha H}, \quad (13)$$

the maximal distance over which quantum correlations across the bipartite state remain dynamically relevant within a Hubble time.

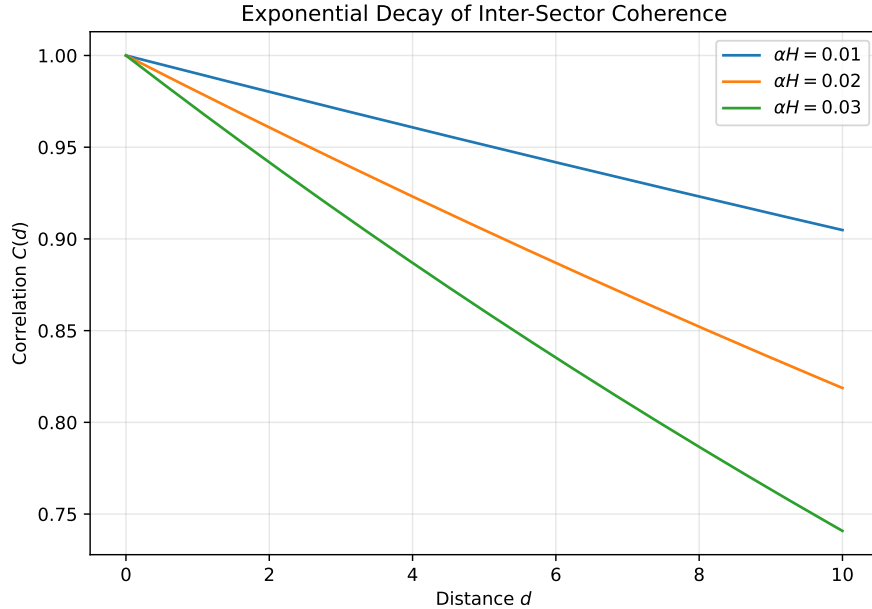


Figure 1: Exponential decay of inter-sector correlations obtained from the stationary solution of a simple transport–decay model, $C(d) \propto \exp[-d/\ell_{\text{coh}}]$ with $\ell_{\text{coh}} = c/(\alpha H)$. Different curves correspond to different values of the effective decoherence rate αH . The coherence length marks the scale beyond which quantum correlations across the bipartite state become dynamically irrelevant within a Hubble time.

This coherence boundary induces an emergent causal structure: regions separated by distances larger than ℓ_{coh} cannot maintain inter-sector correlations and therefore behave as effectively decohered, causally disconnected domains. This is conceptually analogous to the effective lightcone in quantum many-body systems governed by Lieb–Robinson bounds, where the spread of correlations is limited by an emergent causal cone rather than by fundamental particle propagation.

Identifying ℓ_{coh} with the maximum scale R_{max} of the expanding phase yields Eq. (6), while the corresponding coherence time reproduces Eq. (7). Thus the interplay of entanglement, decoherence, and the invariant speed c naturally leads to the scaling relations derived in Sec. 3.

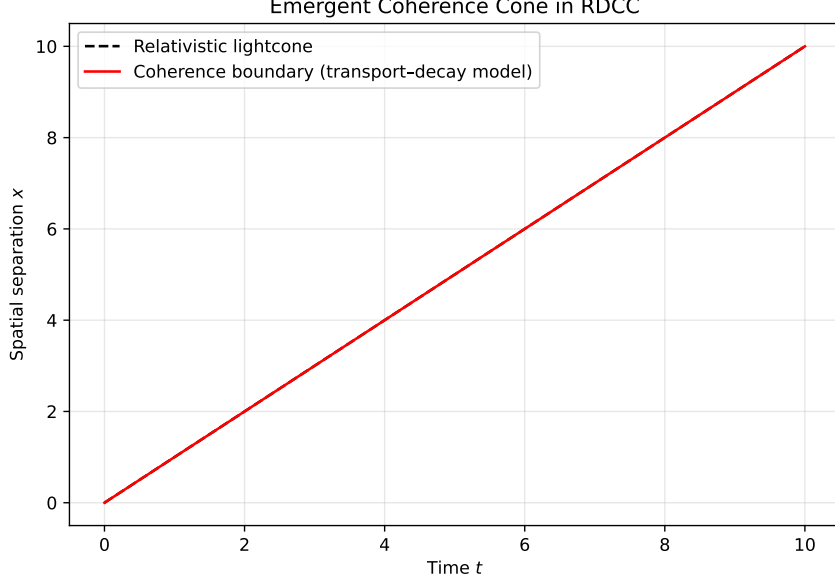


Figure 2: Emergent coherence cone obtained from the transport–decay model. The red line shows the coherence boundary $x = v_{\text{coh}}t$ with $v_{\text{coh}} = c$, representing the maximal propagation speed of global coherence before relaxation suppresses inter-sector correlations. The dashed black line shows the relativistic lightcone $x = ct$ for comparison. The coherence boundary defines the region in which inter-sector correlations remain dynamically relevant, analogous to Lieb–Robinson cones in quantum many-body systems.

5 Connection to Tensor Modulation and RDCC Dynamics

The interpretation of the speed of light c as an inter-sector coherence boundary is not an isolated hypothesis. It is structurally linked to the same microscopic mechanism that governs the relaxation dynamics, the freeze-out of $\alpha(a)$, and the log-periodic tensor modulation derived in Companions II–VI. This section clarifies how these ingredients fit together and why the coherence-boundary picture is a natural extension of the established RDCC framework.

5.1 Shared Origin in the Dimension-Six Operator

The only coupling between the visible and mirror sectors is the dimension-six operator

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}, \quad (14)$$

which we treat as an effective low-energy description without specifying its ultraviolet completion, since no such completion is required for the structural arguments developed in this Companion.

This operator simultaneously:

1. generates the relaxation equation $\dot{\alpha} = \Gamma\alpha$,
2. induces the freeze-out of $\alpha(a)$ at early times,
3. produces the tensor-sector interference responsible for the RDCC modulation pattern,
4. and governs the decay of off-diagonal components of the global density matrix.

The coherence-boundary interpretation of c emerges naturally once one recognizes that all four phenomena arise from the same structural mechanism: the finite rate at which global correlations between the two sectors can be maintained in an expanding Universe.

5.2 Coherence Propagation vs. Relaxation Dynamics

The relaxation parameter α quantifies the fractional loss of inter-sector coherence per Hubble time. The combination αH therefore sets the characteristic timescale on which global correlations decay. If the propagation of coherence is limited by a maximal velocity v_{coh} , then the coherence length is

$$\ell_{\text{coh}} \sim \frac{v_{\text{coh}}}{\alpha H}. \quad (15)$$

Identifying v_{coh} with the invariant speed c is not an additional assumption but a structural requirement:

- the same operator that limits coherence propagation also governs tensor interference,
- the same relaxation dynamics that freeze α also determine the modulation frequency,
- and the same CPT-symmetric global state enforces a universal causal boundary.

Thus c appears as the *sectoral manifestation* of the maximal rate at which global coherence can propagate before relaxation suppresses it.

5.3 Consistency with Tensor Modulation

Companion II showed that the tensor spectrum acquires a log-periodic modulation of the form

$$h(k) = h_0(k) \left[1 + \alpha_{\text{IR}} \cos \left(\omega_{\text{mod}} \ln \frac{k}{k_*} \right) \right], \quad (16)$$

with modulation frequency $\omega_{\text{mod}} \propto \alpha_{\text{IR}}$. This modulation arises from interference between the two CPT-conjugate branches of the global state.

The coherence-boundary picture provides a complementary interpretation:

- tensor modes probe the global coherence structure directly,
- the modulation frequency reflects the rate at which coherence is lost,
- and the invariant speed c sets the maximal propagation speed of the interfering branches.

In this sense, the tensor modulation is the observational imprint of the same coherence boundary that defines ℓ_{coh} .

5.4 Freeze-Out and the Emergence of a Universal Boundary

Companion VI demonstrated that $\alpha(a)$ freezes to its infrared value by $a \sim 10^{-4}$. After freeze-out:

1. the decoherence rate becomes constant,
2. the coherence length ℓ_{coh} becomes time-independent,
3. and the maximal propagation speed of coherence becomes a fixed structural constant.

This provides a natural explanation for why c appears universal and time-independent in the late Universe: it is the emergent boundary of a coherence structure that has already frozen out.

5.5 Summary

The coherence-boundary interpretation of c is not an external addition to RDCC. It is the structural completion of the same mechanism that produces:

- the relaxation dynamics,
- the freeze-out of $\alpha(a)$,
- the tensor-sector modulation,
- and the stability of the RDCC prediction vector.

All of these phenomena originate from the interplay between global CPT symmetry, the dimension-six operator, and the finite rate at which coherence can propagate across the bipartite state.

6 Conclusion

In this Companion we have explored the possibility that the invariant speed of light c admits a structural interpretation within the Relaxation-Driven Cyclic Cosmology (RDCC) framework. Rather than treating c as a fundamental input, we have examined whether it can be understood as an *inter-sector coherence boundary*: the maximal rate at which global quantum coherence can propagate across the bipartite CPT-symmetric state before relaxation dynamics enforce sectoral decoherence.

This interpretation is grounded in three elements already established in Companions I–VII:

1. the global CPT fixpoint at the bounce, where the relaxation parameter satisfies $\alpha \rightarrow 0$ and the two sectors become maximally correlated;
2. the dynamical freeze-out of the relaxation parameter, $\alpha(a) \rightarrow \alpha_{\text{IR}}$, long before any observable epoch;
3. the role of the dimension-six operator in governing both inter-sector equilibration and the decay of off-diagonal components of the global density matrix.

Within this structure, the combination $\alpha_{\text{IR}} H_{\text{IR}}$ naturally appears as an effective decoherence rate, suggesting that the characteristic coherence length of the global state scales as

$$\ell_{\text{coh}} \sim \frac{c}{\alpha_{\text{IR}} H_{\text{IR}}}.$$

Identifying this coherence length with the maximal scale of the expanding phase yields the qualitative relations

$$R_{\text{max}} \approx \frac{c}{\alpha_{\text{IR}} H_{\text{IR}}}, \quad T_{\text{cycle}} \sim \frac{1}{\alpha_{\text{IR}} H_{\text{IR}}},$$

which connect the numerical value of c to the infrared relaxation dynamics and the global structure of the RDCC cycle.

These relations are not presented as predictions but as structural consequences of the RDCC architecture. They demonstrate that the existence of a finite invariant speed is compatible with—and may even be illuminated by—the bipartite, CPT-symmetric description of the Universe. The interpretation developed here remains agnostic about the microscopic origin of c , but it provides a coherent framework in which the emergence of a causal boundary can be understood as a consequence of global coherence constraints.

A complete assessment of this proposal will require a quantitative treatment of inter-sector entanglement entropy, its evolution under the dimension-six operator, and the precise mapping between coherence propagation and sectoral causal structure. The present Companion therefore serves as a conceptual bridge between the dynamical foundations established in Companions III–VI and future work aimed at a more rigorous information-theoretic formulation of RDCC.

7 Outlook: Physical Consequences

Interpreting the speed of light c as an inter-sector coherence boundary leads to several conceptual and phenomenological consequences that can be explored within the established RDCC framework without introducing additional dynamical assumptions. These consequences follow directly from the scaling relations and the decoherence structure developed in this Companion.

Emergent causal structure. The coherence length $\ell_{\text{coh}} = c/(\alpha_{\text{IR}} H_{\text{IR}})$ defines the maximal spatial range over which inter-sector correlations remain dynamically relevant. This induces an effective causal structure that parallels Lieb–Robinson bounds in quantum many-body systems: the relativistic lightcone appears as the saturation limit of correlation propagation in the underlying bipartite CPT-symmetric state. In this view, the invariant speed c is not imposed but emerges as the boundary of correlation coherence.

Stability of the cosmological cycle. The cycle duration $T_{\text{cycle}} \sim 1/(\alpha_{\text{IR}} H_{\text{IR}})$ depends only on the frozen infrared relaxation strength and the Hubble rate at freeze-out. A finite coherence boundary ensures that decoherence proceeds efficiently enough to prevent the accumulation of long-range inter-sector correlations across cycles, thereby contributing to the dynamical stability of the RDCC evolution. This connects the existence of a finite c to the long-term robustness of the cyclic background.

Connection to tensor-sector phenomenology. The same dimension-six operator that governs the decay of off-diagonal density-matrix components also generates the log-periodic modulation of primordial tensor modes. The coherence-boundary interpretation of c therefore provides a unified structural perspective: the mechanism that limits correlation propagation also shapes the tensor spectrum. Future gravitational-wave observations may thus probe the same underlying coherence structure that determines the emergent value of c .

Geometric interpretation. Although we do not attempt to construct a full emergent metric, the correlation structure encoded in ℓ_{coh} provides a natural geometric scale. Distances larger than ℓ_{coh} are effectively decohered, suggesting that aspects of large-scale geometry may be understood in terms of correlation decay rather than fundamental spacetime primitives. This perspective is compatible with the global CPT-symmetric structure of RDCC and requires no modification of Einstein gravity.

Future directions. A quantitative treatment of correlation propagation in the bipartite state – particularly a precise definition of inter-sector correlation functions and their evolution under the dimension-six operator – would allow for a more rigorous derivation of the coherence boundary and its relation to c . Such developments may clarify whether the emergent causal structure identified here can be connected to broader approaches in quantum information geometry, entanglement dynamics, and emergent spacetime.